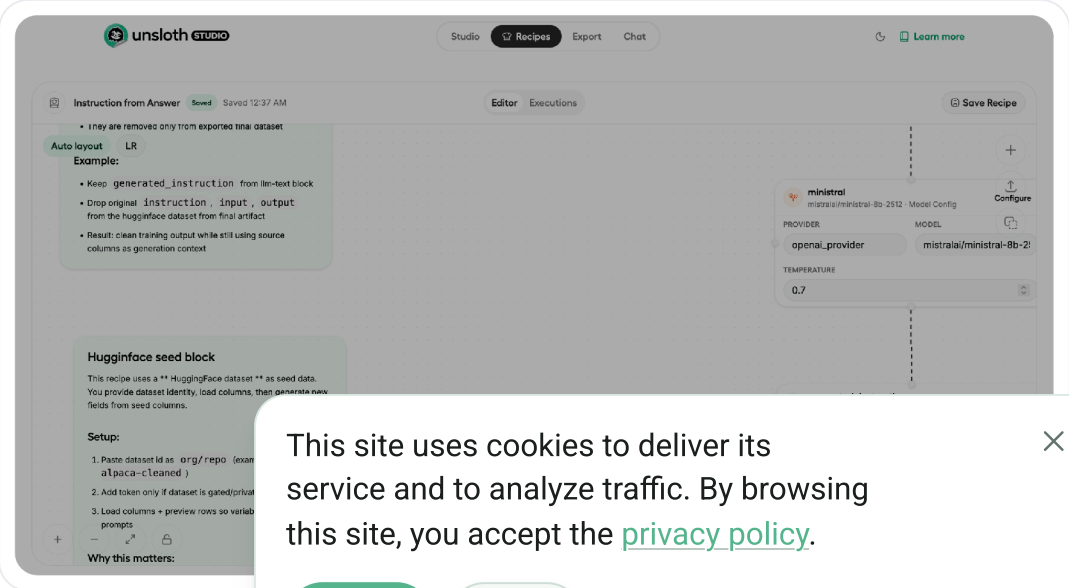
Unsloth Data Recipes

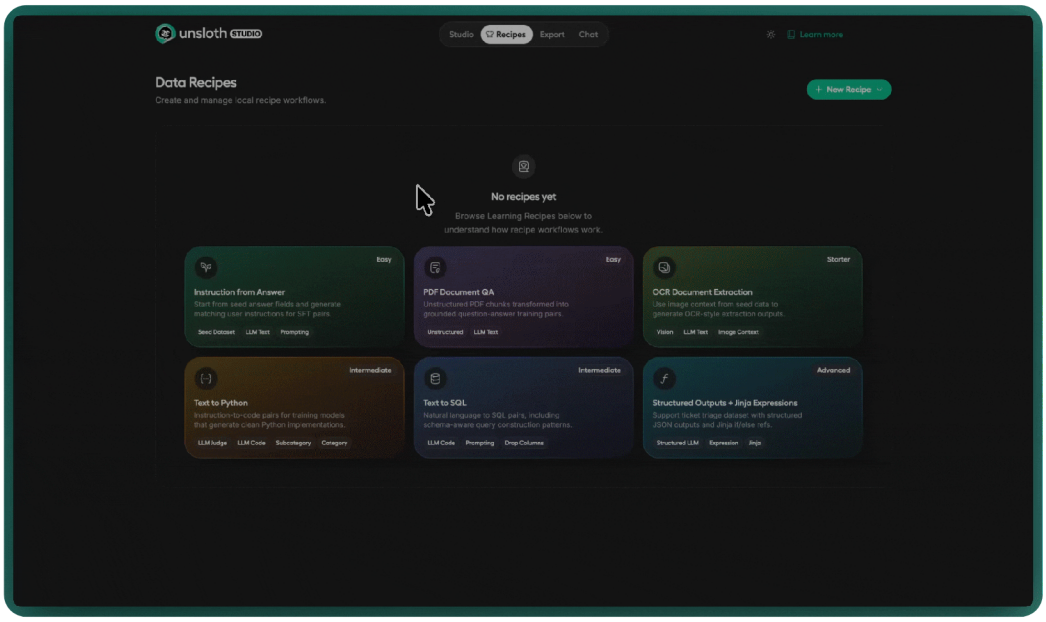
Learn how to create, build and edit datasets with Unsloth Studio's Data Recipes.

Unsloth Studio's Data Recipes lets you upload documents like PDFs or CSVs files and transforms them into useable / synthetic datasets. Create and edit datasets visually via a graph-node workflow. This guide will get you started with the basics before you dive into Unsloth Data Recipes.



How Data Recipes works

Data Recipes follows the same basic path. You open the recipes page, create or pick a recipe, build the workflow in the editor, validate it run a preview, then run the full dataset once the output looks right. Add seed data and generation blocks, validate the workflow, preview sample output, then run a full dataset build. Unsloth Data Recipes is powered by **NVIDIA Nemo** Data Designer ↗.



Example of generating dataset and fine-tuning a model

At a glance a usual workflow should look like this:

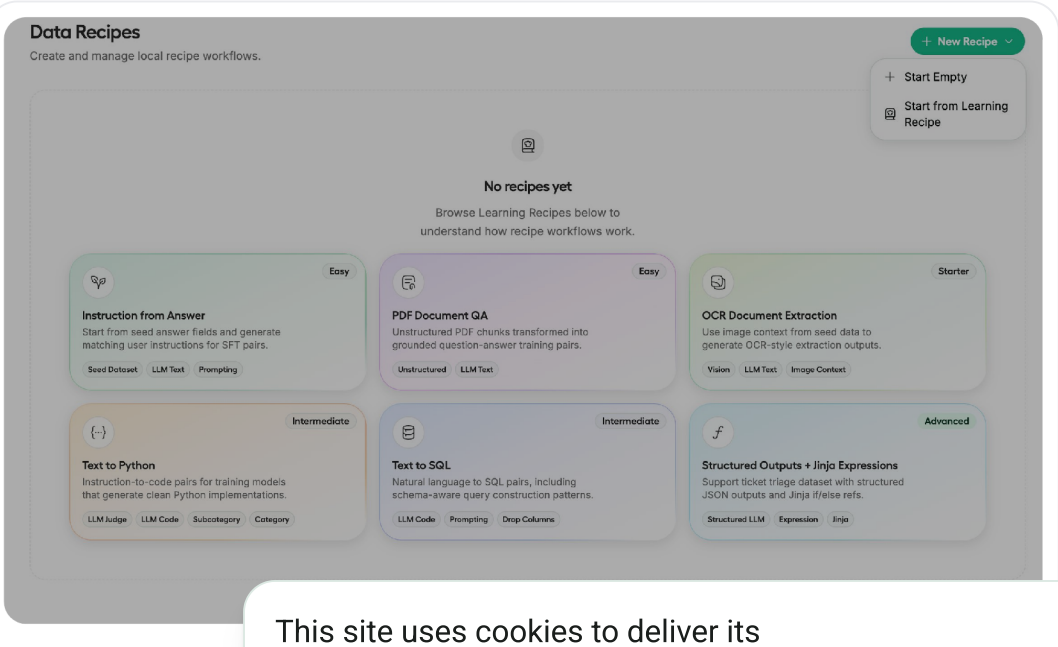
1. Open the recipes page.
2. Create a new
3. Add blocks t
4. Click **Validat**
5. Run a previe
6. Run a full dataset build when the recipe is ready.

7. Review progress and output live in graph or in **Executions** view for mode details.
8. Select the resulting dataset in **Unsloth** and fine tune a model.

Get Started

The recipes page is the main entry point. Recipes are stored locally in the browser, so you come back to saved work later. From here, you can create a blank recipe or open a guided learning recipe.

> Recipes can be exported and imported, so it is easy to share workflows with other Unsloth users 🎉. If you are trying to build a specific dataset pattern, ask in Unsloth Discord. Someone may already have a recipe they can share.



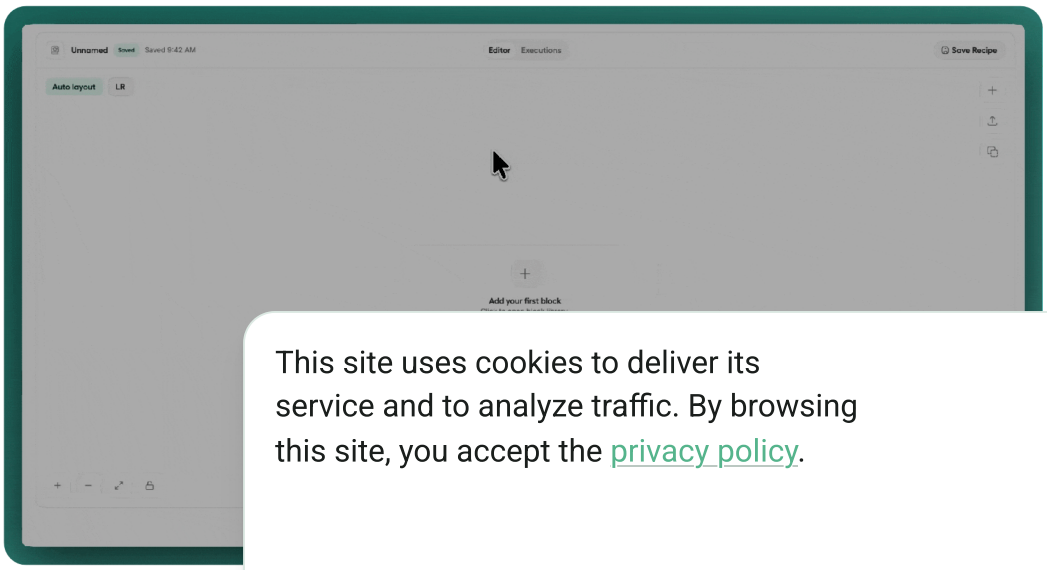
If you are new to concept of workflows, learning recipes are the fastest way to see how seed data, prompts, expressions, and validators fit together in one working example. If you already know the shape of dataset you want, starting empty is usually quicker.

Choose a starting path

If you want to:	Start with:
Build a custom workflow quickly	**Start Empty**
Learn the product from an example	**Start from Learning Recipe**
Continue previous work	**Open a saved recipe**

What you build in the editor

The editor is where the recipe takes shape. You add blocks from the block sheet, configure them in dialogs, connect them on the canvas, and then validate or run the workflow.



Example of building product description workflow

The editor has a few core parts:

- The recipe header, where you rename the recipe and switch between **Editor** and **Executions**
- The canvas, where the recipe graph is shown
- The block sheet, where you add new blocks
- Configuration dialogs, where you define prompts, references, model aliases, validators and seed settings.
- The floating **Run** and **Validate** controls
- need to add more here

The most common blocks in reciper are:

- **Seed** for input data from hugginface, local structured files (or unstructured documents that get chunked into rows.
- **LLM + Models** for providers, model configs, LLM generation blocks, and shared tool profiles.
- **Expression** for jinja2-based transforms that do not require an LLM call.
- **Validators** for filtering bad generated code with built in linters for Python, SQL, and Javascript/Typescript.
- **Samplers** for deterministic columns such as categories and subcategories.

How references work

Most blocks that
reference for lat
Data Recipes. Yo
expressions, stru

Jinja Expressions help you work with values that arleady exist in the recipe. You can reference nested fields like `{{customer.first_name}}` , join values like `{{customer.first_name}} {{customer.last_name}}` and add conditional logic with patterns such as `{% if condition %}...{% endif %}`

Example of references shown in the editor

For example:

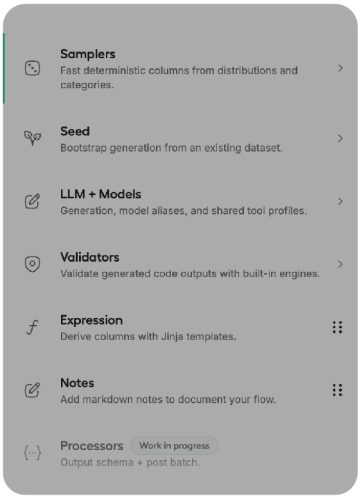
- A category block named `domain` can be references as `{{ domain }}`
- a seed column can be used directly in an LLM prompt, the columns in your seed data (eg. HF dataset columns, csv)
- a structured LLM output can expose fileds for later prompts
- an expression block can combine earlyier values without another model call

What happens after?

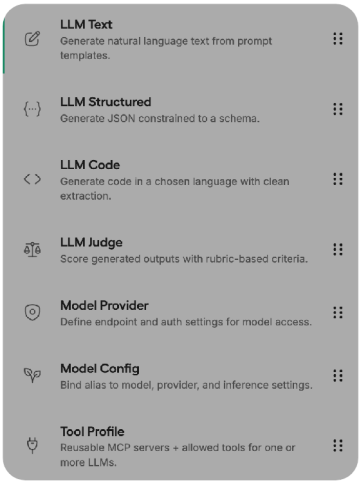
Preview runs are for quick iteration. They return sample rows and analysis in the editor so you can inspect the generated data before commiting to a full run.

Full runs create a
appears in Unslo
again and use it
dataset to you h

Core building blocks



Core building blocks



Model and LLM blocks

Model setup is split into two usable layers:

- **Model provider** defines the endpoint and authentifcation
- **Model Config** defines the model name and inference settings

This setup works with hosted providers, self-hosted endpoints, `vLLM` , `llama.cpp` , or any OpenAI-compatible API that you run outside Unsloth.

> Recipes are not limited to one model. You can add multiple **Model providers** and **Model config** blocks, then use different models for different steps, such as one for coding and another for general text tasks.

After model setu

Block	Output	Best for
LLM Text	Free-form text	Instructions, explanations, conversations, and descriptions
LLM Structured	JSON	Output that need fixed fields and predictable structure
LLM Code	Code	Python, SQL, Typescript and other code generation tasks
LLM Judge	Scored evaluation	Grading outputs with one or more user-defined score

Tool Profiles

Tool profile blocks defines shared MCP based tool access for one or more LLM blocks. Use them when a generation step needs tools, such as looking up code documentation through `Context7`.

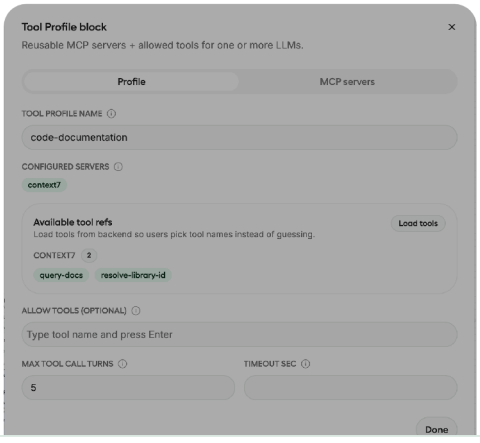


Image to the left
Context7 MCP a
configured in To
dialog:

Validators

Validor block primarly target LLM code block by running generated code outputs through Linter and syntax validation, this helps you keep bad or invalid code rows out of the final dataset by filtering them out. The built-in options cover Python, SQL, and JavaScript/TypeScript validation.

Validate, preview and run

Once the recipe workflow is in place, the next step is execution. The reccomended pattern is: validate first, preview for quick feedback and inspect the generated data in executions view, then run the full dataset when you feel the output satisfies your plan.

Use the execution controls in third order:

1. **Validate**

   Click **Vali**

2. **Preview**

   Run a pre

3 Refine

Refine prompts, references, seed settings, or validators.

Iterate untill you feel satisfied with generated data

4 Run the full dataset build

Previous
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